Sounding Board

Risks and Benefits of Activated Protein C Treatment for Severe Sepsis

A NEW hypothesis with implications for the treatment of sepsis has recently been tested — the hypothesis that part of the pathophysiology of sepsis is caused by unrestricted or inappropriate coagulation in the microcirculation. Three agents that block coagulation at different stages have now been evaluated in large, multicenter trials as adjunctive therapy for sepsis. Neither antithrombin III nor tissue-factor-pathway inhibitor was effective in patients with severe sepsis or septic shock.[1,2] In contrast, treatment with activated protein C (drotrecogin alfa [activated]) was associated with a significant reduction in 28-day mortality. Of the patients who received activated protein C, 24.7 percent had died at 28 days, as compared with 30.8 percent of those who received placebo ($P=0.005$).[3] However, the use of activated protein C was also associated with greater risk of serious bleeding (3.5 percent vs. 2.0 percent, $P=0.06$).

In November 2001, the Food and Drug Administration (FDA) approved recombinant human activated protein C (Xigris, Eli Lilly) for the treatment of patients with severe sepsis who have a high risk of death (as determined, for example, by the Acute Physiology and Chronic Health Evaluation [APACHE II] severity-of-illness scoring system). The purpose of this article is to provide clinicians with information that is not found in the published account of the clinical trial of activated protein C or in the marketing literature prepared by the sponsor. Most of this information was presented by the FDA at the October 16, 2001, meeting of the Anti-Infective Drugs Advisory Committee.[4] A more detailed account of this meeting is available on the Internet.[5]

Three important issues regarding the clinical use of activated protein C arose during these deliberations. First, there were changes in the study protocol and in the drug preparation that appeared to divide the trial into two discernible phases in which the efficacy of activated protein C was different. Second, there was new information about the efficacy of the drug in various subgroups of patients. Third, there were new data on the toxicity of the drug.

THE TWO PHASES OF THE ACTIVATED PROTEIN C TRIAL

In June 1999, after 720 patients had been enrolled in the trial of activated protein C, the sponsor amended the study protocol. The entry criteria were modified to exclude patients who had received bone marrow transplants or solid-organ transplants, who had metastatic cancer or pancreatitis, or for whom a commitment for aggressive treatment had not been obtained. Patients who had had organ failure for more than 24 hours at the time when they met all other criteria for inclusion were also excluded. These changes, which were made in a blinded manner and before the first interim analysis by the data and safety monitoring committee, shifted the composition of the study population toward patients with less severe underlying disease and more acute infectious illnesses. In addition, a new placebo (0.1 percent albumin instead of saline) was introduced, and protein C deficiency status was eliminated as a primary variable for the analysis. The manufacture of activated protein C requires expression of the recombinant protein from cells in culture, enzymatic cleavage of the protein, and several other post-translational modifications.[6] In August 1999, a new master lot of cells was introduced to make recombinant activated protein C. Extensive in vitro studies did not reveal differences between the old and new preparations of activated protein C. The new preparation was used for the remainder of the trial.

After these changes were made, there was an improvement in the protective efficacy of activated protein C. The cumulative mortality curves for the trial are shown in Figure 1. Because these curves depict mortality at 28 days and are graphed according to the date of randomization and because of variable phase-in times, the effects of changes are likely to be apparent after a slight delay. For most of the trial, mortality in the placebo group was relatively flat. The mortality in the activated protein C group decreased somewhat beginning in February 1999, then increased, and then decreased progressively beginning in October 1999. The FDA analysis indicated that the efficacy of the drug after the amendment of the protocol differed substantially from its efficacy before the changes were made. Whereas activated protein C did not improve survival in the phase before amendment of the protocol (720 patients; mortality in the placebo group, 30 percent; mortality in the activated protein C group, 28 percent; relative risk of death, 0.94; $P=0.57$), it was very effective in the phase after the protocol was amended (970 patients; mortality in the placebo group, 31 percent; mortality in the activated protein C group, 22 percent; relative risk of death, 0.71; $P=0.001$). It is not possible to tell from the available data whether the striking improvement in efficacy resulted from the enrollment of a somewhat different population of patients, an undetected change in the drug itself, chance alone, or some combination of the three.

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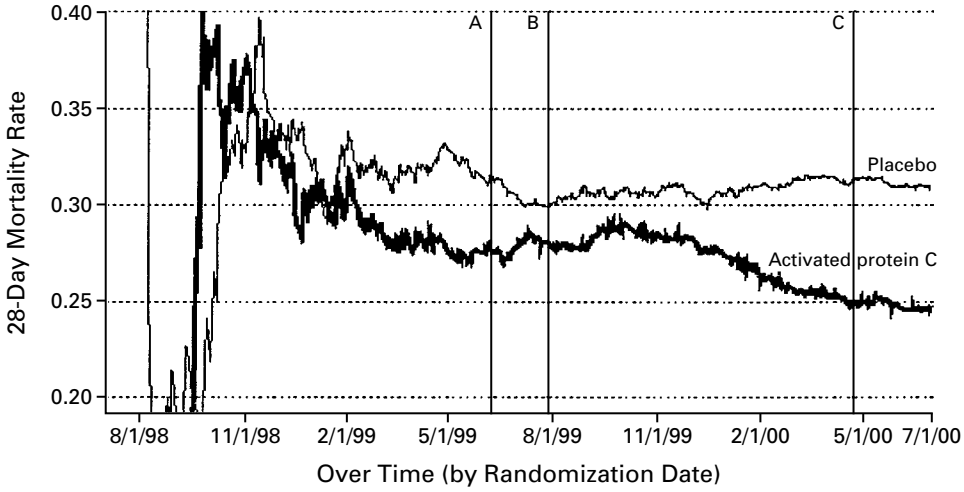


Figure 1. 28-Day Cumulative Mortality over Time for All Patients.

The amended version of the protocol was introduced at the time indicated by line A. The first interim analysis occurred at the time indicated by line B. The change in the manufacturing of the drug also occurred at approximately this time. The second interim analysis occurred at the time indicated by line C. The graph is from the FDA.[4]

SUBGROUPS AND RESTRICTION TO USE IN HIGH-RISK PATIENTS

Although the trial had the power to determine whether activated protein C is more effective than placebo in patients with severe sepsis, the efficacy of the drug was also examined in numerous subgroups. Definitive conclusions often cannot be reached on the basis of post hoc analyses of subgroups within a large clinical trial, especially if the trial does not have sufficient power to detect differences in smaller numbers of patients. Nevertheless, activated protein C seems to have been effective in numerous subgroups. Its efficacy was most apparent in patients older than 50 years of age, in patients with more than one dysfunctional organ system, in patients with an APACHE II score of more than 24 before the infusion of the drug, and in patients who had shock at the time of the infusion. Subgroups in which benefit was not apparent included patients who had undergone surgery and patients with failure of a single organ.

The FDA licensed activated protein C for the treatment of adult patients with severe sepsis who have a high risk of death, as indicated by an APACHE II score of 25 or more. APACHE II is a bedside scoring system that combines physiological data with information about age and chronic disease.[7] It was developed to predict a patient's risk of dying in the intensive care unit and was designed and validated for patients who have been in the intensive care unit for 24 hours. In the activated protein C trial, APACHE II scores before the infusion of activated protein C were calculated as a measure of the severity of illness, in part to ensure comparability of the patients enrolled in the two treatment groups. Data were collected for the calculation of the APACHE II score within the 24-hour period preceding the administration of the study drug. The FDA recommendation was based on the finding in this pivotal trial that activated protein C was most beneficial in patients whose APACHE II score before infusion exceeded 24.

There is, however, no precedent for using the APACHE II system to select patients for novel therapies. When it is applied to the administration of activated protein C in clinical practice, the APACHE II score will most likely be based on data acquired in "real time." A score calculated in this way may change as the patient's physiological status improves or deteriorates. It is unclear whether or not patients should receive activated protein C if their APACHE II score falls from, say, 26 to 23 as they are resuscitated before the infusion of the drug has begun. Furthermore, the interobserver and intraobserver variability in the determination of APACHE II scores among experienced intensive care physicians may be as high as 10 to 20 percent.[8,9] The use of APACHE II scores in this new manner — as a screening test to determine which patients should receive a new therapy and which patients should not — has not been validated. Correct use of APACHE II scoring is of major importance, because activated protein C was not effective in patients who had APACHE II scores of less than 20, and it was associated with an increased risk of bleeding in such patients (Table 1).

An additional consideration for practitioners may be

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TABLE 1. MORTALITY AND BLEEDING ACCORDING TO APACHE II QUARTILE.*

APACHE II QUARTILE (SCORE)	28-DAY MORTALITY		RELATIVE RISK (95% CI)	SERIOUS BLEEDING		RELATIVE RISK (95% CI)†
	APC	PLACEBO		APC	PLACEBO	
	percent			percent		
1st (3–19)	15	12	1.25 (0.78–2.02)	4	0	18.3 (2.7–)† P=0.003
2nd (20–24)	23	26	0.88 (0.63–1.22)	1	2	0.5 (0.004–6.1) P=1.0
3rd (25–29)	24	36	0.66 (0.48–0.91)	3	0	10.3 (1.4–)† P=0.02
4th (30–53)	38	49	0.78 (0.65–0.96)	1	1	1.1 (0.04–32.2) P=1.0

*APACHE II denotes Acute Physiology and Chronic Health Evaluation, CI confidence interval, and APC activated protein C. Serious bleeding consisted of intracranial hemorrhage, a life-threatening episode of bleeding, or the requirement for 3 or more units of blood per day for two consecutive days. Data are from the briefing document of the FDA Anti-Infective Advisory Committee.[4]

†The upper 95 percent confidence limit was a very large number.

the fact that, despite the broad entry criteria used for the study, the patients who were actually enrolled in the trial represented a rather narrow subgroup of the total population of patients with sepsis found in most medical centers. More than 80 percent of the patients in the trial were living at home before hospitalization, and more than 50 percent were admitted with respiratory failure. Patients who had had organ failure for more than 24 hours were excluded. The risk–benefit ratio of the drug in the patients studied might well differ from that in patients who have been in the hospital for a much longer time, yet the latter group of patients might still meet the criteria established by the FDA for treatment with the drug.

RISK OF SERIOUS HEMORRHAGE

The use of activated protein C was associated with an increased risk of serious bleeding in the controlled, randomized trial. Four deaths due to hemorrhage occurred during the infusion of activated protein C, whereas none occurred during the infusion of placebo. A total of 3.5 percent of patients who received activated protein C had serious bleeding (defined as intracranial hemorrhage, a life-threatening bleeding episode, or a requirement for 3 or more units of blood per day for two consecutive days, or any bleeding episode otherwise assessed as serious), as compared with 2 percent of the patients who received placebo ($P=0.06$). The incidence of serious bleeding may increase when activated protein C is used in less controlled circumstances. During the open-label use of activated protein C after the trial, for which similar criteria were used for the selection of patients in order to minimize the risk of bleeding, 13 of 520 patients, or 2.5 percent, had an intracranial hemorrhage, and eight of these hemorrhages occurred during the infusion period.[10] Only 0.2 percent of the patients who received the drug in the controlled trial had this complication.

CONCLUSIONS

Activated protein C is the first biologic agent approved in the United States for the treatment of severe sepsis. Its success came after almost two decades of intensive research and more than 20 large clinical trials of other candidate therapeutics. As is the case with other anticoagulants, it may cause serious hemorrhage, and the risk of this complication needs to be assessed in the context of the potential benefit of the drug.

The overriding task that now faces clinicians is to decide which patients should receive activated protein C and whether the balance of available data supports its widespread use. For many drugs, extensive testing in animals provides reassurance that the concept, mechanisms of action, and drug formulation are compatible with the clinical results. In this case, the proposed mechanisms of protection by activated protein C are unproven. In the pivotal trial, there was no relation between protein C levels before infusion or measures of clotting after infusion and the efficacy of the drug; moreover, the antiinflammatory properties proposed for activated protein C require confirmation in vivo. Unfortunately, the species specificity of activated protein C has limited its testing as a treatment for sepsis to studies in baboons in which a very large bacterial challenge is used.[11] Interpretation of the single trial involving human subjects thus has special importance.

Several important points support the conclusion that activated protein C is effective in at least some pa-

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tients with severe sepsis. In particular, activated protein C is the first antisepsis drug that has met predefined efficacy criteria for premature termination of a trial, and the level of statistical significance reached was impressive ($P=0.005$). In addition, the drug was efficacious in many different subgroups. There are, however, some points of concern. Many of these derive from the fact that populations of patients with sepsis are highly heterogeneous, so that trials enrolling patients with sepsis are difficult to perform, interpret, and reproduce.[12,13] Careful analysis failed to reveal a reason why the efficacy of activated protein C was not more consistent throughout the trial. It is impossible to exclude the possibility that, in some way, the amendments to the protocol or the changes in the drug may have been responsible. Of particular concern is the possibility that the significant results in the second phase of the trial may reflect the use of a functionally different lot of the protein and that there is no laboratory test that can predict the efficacy of future lots. The difficulty in balancing these conflicting ideas was reflected by the vote of the FDA Anti-Infective Drugs Advisory Committee, which was split 10 to 10 as to whether activated protein C is safe and efficacious.

Although the data regarding activated protein C are very encouraging, we believe that there is not sufficient evidence at present for it to become the standard of care. Although the data from the trial suggest that activated protein C was efficacious in a subgroup of patients with high APACHE II scores before infusion, this hypothesis needs to be tested in a confirmatory trial that prospectively incorporates a prognostic scoring system such as APACHE II. A successful outcome of such a trial would constitute a major advance, would settle the current controversy, and would justify the use of the drug in this subgroup of patients with severe sepsis.

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Assessing the Use of Activated Protein C in the Treatment of Severe Sepsis

In 2001, the Food and Drug Administration (FDA) evaluated an application for the use of drotrecogin alfa (activated), or recombinant human activated protein C (Xigris, Eli Lilly), in patients with severe sepsis. The use of activated protein C, as compared with placebo, was associated with a significant reduction in mortality in the Recombinant Human Activated Protein C Worldwide Evaluation in Severe Sepsis (PROWESS) trial (24.7 percent vs. 30.8 percent, $P=0.005$).[1] Extensive review by physicians and scientists at the FDA confirmed the principal findings of the PROWESS trial but raised issues regarding the interpretation of data and appropriate use of the drug.

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These issues were discussed with experts on sepsis from outside the FDA at a public meeting of the agency's Anti-Infective Drugs Advisory Committee on October 16, 2001.[2,3] The committee members provided valuable input; highlighted specific issues of concern, some of which are summarized in the accompanying Sounding Board by Warren et al.[4]; and were evenly divided in a vote on a recommendation regarding approval. The FDA then performed substantial additional evaluations addressing the issues that had been raised and concluded that activated protein C had been demonstrated to be safe and effective in reducing mortality among patients with severe sepsis and a high risk of death, as determined, for example, by the Acute Physiology and Chronic Health Evaluation (APACHE II) score.[5] In this article, I summarize the findings of the FDA regarding several key matters of concern — in particular, changes made during the trial, the use of APACHE II scores, and the risk of serious bleeding.

CHANGES DURING THE TRIAL

Near the midpoint of the activated protein C trial, the study protocol was amended, and activated protein C produced with the use of a new master cell bank was introduced.[2,3] The reduction in mortality associated with treatment with activated protein C was greater after these changes were made than it had been up to that point (Table 1), raising questions about whether changes in the protocol, the drug, or both accounted for the improved results and therefore about the applicability of the results of the trial to the use of activated protein C.[2,3] The apparent inconsistencies over the course of the study were limited to the patients with lower risk; the high-risk population (patients with an APACHE II score of 25 or more), covered by the indication approved by the FDA, showed consistent benefit (Table 1).

FDA analyses indicate that differences over time as large as or larger than those seen in the PROWESS study would be expected to occur by chance alone in 8 percent of trials. As summarized below, FDA analyses do not support the belief that changes in either the protocol or the drug accounted for such differences.[6]

Patients with Serious Preexisting Diseases

In order to increase the capacity of the trial to detect effects on mortality due to sepsis, the eligibility criteria for the activated protein C study attempted to exclude patients who were likely to die from underlying disease within 28 days. The amendments to the eligibility criteria were intended to make this exclusion more effective. Two analyses indicate that these amendments did not account for the improved study outcomes.

First, among the patients enrolled before the protocol was amended, those who would not have met the amended eligibility criteria actually had a substantially larger absolute reduction in mortality (8 percent) (Table 1) than those whose eligibility would not have been affected by the changes (1 percent).[2,3,6] Second, in analyzing the effect of underlying disease on treatment effect, the FDA used the chronic health points that had been prospectively assigned to subjects. Chronic health points, a component of the APACHE II score that assesses the risk of death,[5] are assigned only to those patients with preexisting severe organ failure or immunosuppression. It was just

TABLE 1. OUTCOME DATA FROM THE PROWESS TRIAL BEFORE AND AFTER AMENDMENT OF THE PROTOCOL.

GROUP OF PATIENTS	MORTALITY BEFORE AMENDMENT			MORTALITY AFTER AMENDMENT		
	PLACEBO GROUP	ACTIVATED PROTEIN C GROUP	ABSOLUTE DIFFERENCE*	PLACEBO GROUP	ACTIVATED PROTEIN C GROUP	ABSOLUTE DIFFERENCE*
	no./total no. (%)		%	no./total no. (%)		%
All patients	109/360 (30)	102/360 (28)	−2	150/480 (31)	108/490 (22)	−9
APACHE II score						
<25	34/185 (18)	48/195 (25)	7	49/252 (19)	34/241 (14)	−5
≥25	75/175 (43)	54/165 (33)	−10	101/228 (44)	74/249 (30)	−15
Amended eligibility criteria						
Would not be met	17/40 (42)	14/41 (34)	−8	0/0	0/0	
Would be met	92/320 (29)	88/319 (28)	−1	150/480 (31)	108/490 (22)	−9

*The difference is shown as the mortality in the activated protein C group minus the mortality in the placebo group.

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such patients in whom most of the beneficial effect of activated protein C therapy was seen (Table 2). Treatment with activated protein C was associated with a 3 percent absolute reduction in mortality among patients without chronic health points on APACHE II but with a 19 percent absolute reduction in mortality among patients who had chronic health points. The P value for the interaction of treatment effect and APACHE II chronic health points was 0.01.

Both of these analyses indicate that the amendments to the protocol, rather than accounting for improved results, actually excluded patients who appeared more likely to benefit from therapy. In addition, these two analyses address the generalizability of the results of the PROWESS trial to patients with sepsis who have serious preexisting disease.[3] They suggest strongly that serious preexisting disease should not, in general, be a reason for withholding activated protein C therapy.

End Points

Notwithstanding some confusion at the meeting of the Anti-Infective Drugs Advisory Committee, resulting from differences between the study objectives and the primary end point,[3] the PROWESS trial had a single, primary, prospectively identified end point — 28-day mortality among all treated patients. This end point did not change during the trial, except that the amended protocol added the protein C activity level as a covariate for adjustment of the primary analysis and eliminated shock as a covariate. This change had a negligible effect on the size of the resulting treatment effect and the associated P value (P=0.008 according to the analysis described in the unamended protocol).[6] The assessment of mortality in subgroups defined according to base-line protein C activity was, from the start, intended to be a secondary analysis. In addition and appropriately, the amendments to the protocol modified the stated study objectives to align them better with the primary end point.

Changes in the Product

Around the time the amendments were made to the protocol, Eli Lilly began using activated protein C that had been produced with the use of a new master cell bank. Changes in a master cell bank can result in changes in the product. The FDA and Eli Lilly conducted exhaustive comparisons of the primary, secondary, tertiary, and quaternary structure, pharmacokinetics, enzyme kinetics, and binding affinity of activated protein C produced from the two master cell banks. No significant differences were found.

Although differences in complex proteins may escape detection, the results of the extensive comparability testing performed indicate that it is unlikely that undetected differences account for the apparent change in clinical outcomes. If the change in the master cell bank did account for the improved clinical results, then the newer, marketed product is superior to the earlier product and highly effective, and one should be wary about future changes in the master

TABLE 2. OUTCOME DATA FROM THE PROWESS TRIAL FOR SELECTED SUBGROUPS.

VARIABLE	MORTALITY IN THE PLACEBO GROUP	MORTALITY IN THE ACTIVATED PROTEIN C GROUP	ABSOLUTE DIFFERENCE*
	no./total no. (%)		%
All patients	259/840 (31)	210/850 (25)	−6
Chronic health points on APACHE II†			
Yes	83/176 (47)	47/169 (28)	−19
No	176/664 (27)	163/681 (24)	−3
Quartile of APACHE II score‡			
First	26/215 (12)	33/218 (15)	3
Second	57/222 (26)	49/218 (22)	−4
Third	58/162 (36)	48/204 (24)	−12
Fourth	118/241 (49)	80/210 (38)	−11
First and second	83/437 (19)	82/436 (19)	0
Third and fourth	176/403 (44)	128/414 (31)	−13

*The difference is shown as the mortality in the activated protein C group minus the mortality in the placebo group.

†Chronic health points are assigned to patients with preexisting severe organ dysfunction, immunosuppression, or both.

‡The ranges of scores in the quartiles were as follows: first quartile, 3 to 19; second quartile, 20 to 24; third quartile, 25 to 29; fourth quartile, 30 to 53.

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cell bank. Eli Lilly has sufficient stores of the current master cell bank to cover at least several decades of activated protein C production (Brophy G, Eli Lilly: personal communication).

USE OF THE APACHE II SCORE

Consideration of measures of risk and severity in general, and the APACHE II score in particular, played an important part in the assessment of the activated protein C trial and in the labeling of activated protein C. The evidence suggesting that the treatment effect of activated protein C varies with the risk of death is strong. Subgroup analyses are most persuasive when they are based on prespecified analyses, have strong biologic plausibility, and are supported by strong and consistent evidence. All these conditions were met in this case.

Analysis of the treatment effect according to the quartile of APACHE II score was prespecified in the PROWESS trial as an important analysis and the principal analysis of outcome according to measures of severity or risk.[2,3] The hypothesis that a treatment effect might be observed best in higher-risk patients with sepsis is a highly plausible one that has been incorporated into many trials. If a drug prevents a fixed proportion of deaths due to sepsis, benefits will be greatest in those populations with the highest risk of death, whereas the frequency of adverse effects might be similar in all subgroups.

In the activated protein C trial, treatment-related benefit consistently increased with the risk of death as assessed by a variety of indexes of base-line severity, including the number of failing organ systems and the presence or absence of shock. Indeed, each of the three risk-related components of the APACHE II score — acute physiological changes, older age, and the assignment of chronic health points — correlated with a greater treatment effect.[2,6]

There are strong reasons for selecting the APACHE II score from among the various measures of risk and severity as a tool for guiding activated protein C therapy. First, the APACHE II score was by far the best discriminator of the risk of death: mortality in the placebo group was 12 percent in the quartile with the lowest APACHE II score and 49 percent in the quartile with the highest score (Table 2). In addition, the APACHE II score was the best predictor of survival benefit from activated protein C. All benefit was observed among the half of subjects who had APACHE II scores of 25 or more (a 13 percent absolute reduction). The subjects with lower risk showed no benefit from activated protein C, and patients in the lowest APACHE II quartile had somewhat higher mortality with activated protein C than with placebo (Table 2).

In contrast, other measures of risk and severity, while supporting the hypothesis that the level of risk determined the amount of benefit, did not discriminate among levels of risk of death or likelihood of benefit as well as the APACHE II score did. For example, low-risk patients defined as those with failure of one organ system (failure of at least one organ system being required for a diagnosis of severe sepsis and for enrollment in the study) had a mortality rate of 21 percent if they were given placebo, as compared with 12 percent for low-risk patients as defined according to the APACHE II score. The patients with failure of one organ system had a mortality rate of 20 percent if they were treated with activated protein C.[2,3,6]

Another advantage of using the APACHE II score rather than other measures in guiding therapy is that the APACHE II score assesses not just the severity of the acute disease process, but also other risk factors that correlated with greater treatment effect — that is, age and preexisting health status. The practicality of using the APACHE II score to guide therapy was assessed by FDA physicians, who found that the APACHE II score could be determined rapidly with the use of data routinely available for patients with severe sepsis.

APACHE II scores in the activated protein C trial were calculated with the use of the most abnormal laboratory measurements and vital signs obtained during the 24 hours preceding treatment; thus, responses to resuscitation would not affect the score. Although the timing of APACHE II scoring differed somewhat from the timing used in validation studies,[2,5] the results of the trial provide strong evidence that the APACHE II score, as assessed in the trial, is a very successful predictor of the risk of death. In clinical practice, the timing of APACHE II scoring to guide the use of activated protein C need not and probably should not deviate substantially from the timing used in the PROWESS trial.

The FDA recognizes that the hypothesis that the benefit of activated protein C in patients with severe sepsis is limited to high-risk patients (those with an APACHE II score ≥ 25), although strongly supported by the data from the PROWESS trial, has not been proved. To assess better the potential for use of activated protein C in lower-risk patients with severe sepsis, the FDA has requested an additional placebo-controlled study involving such patients, and Eli Lilly has committed itself to conducting such a trial with more than 11,000 patients.

RISK OF BLEEDING AND SURVIVAL BENEFITS

Use of activated protein C in severe sepsis is clearly associated with an increased risk of serious hemorrhage during the infusion period — that is, from the start of the infusion through the day after the discontinuation of the infusion. After this period, the rate of

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hemorrhage in the activated protein C group was essentially the same as that in the placebo group. Serious bleeding events (defined in the trial as any intracranial hemorrhage, any life-threatening bleeding episode, any bleeding event requiring the administration of 3 or more units of packed red cells per day for two consecutive days, or any bleeding event otherwise assessed as serious) occurred during the infusion period in 20 of 850 patients treated with activated protein C (2.4 percent) and 8 of 840 patients given placebo (1.0 percent). Although the incidence of intracranial hemorrhage during the infusion period was 0.2 percent (2 of 850) in the activated protein C group in the PROWESS trial, a higher incidence (1.5 percent [8 of 520]) was observed among patients receiving activated protein C in uncontrolled studies. These findings suggest that hemorrhagic complications of activated protein C therapy may be more common outside of highly controlled and monitored settings. They thus emphasize the importance of selection and monitoring of patients in minimizing the risk of hemorrhage. The FDA-approved labeling for activated protein C provides information regarding its safe use.

The absolute survival benefit observed in the PROWESS trial was 6 percent overall and 13 percent in the high-risk population for whom use of activated protein C has been approved (Table 2). Even when one takes into account the possibility that risks may be somewhat higher in clinical practice than they were in the trial, and that some survivors might have prolonged hospitalizations,[2,3] the survival benefits balance quite favorably against the risk of hemorrhage.

CONCLUSIONS

There is still much to be learned about the optimal use of activated protein C. At the request of the FDA, Eli Lilly has made extensive commitments to conduct controlled trials involving more than 13,000 patients to investigate the use of activated protein C in lower-risk patients with severe sepsis, its use in children with severe sepsis, and the optimal use of low-dose heparin with activated protein C.

It is common and appropriate that biologic therapeutics offering important new benefits are approved at a time when the safety and efficacy of their use as specified in the labeling have been established but when much remains to be learned in order to optimize their use. Etanercept (Enbrel, Immunex), infliximab (Remicade, Centocor), trastuzumab (Herceptin, Genentech), rituximab (Rituxan, IDEC), and abciximab (ReoPro, Lilly) are a few examples of drugs about which much has been learned since they received initial approval. The data currently available for activated protein C strongly support the conclusion that the use of this drug as labeled will save many lives as we gather the information necessary to refine its use further.

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spond quickly and affirmatively, and they need the resulting reconstructive procedure to be fundamental, rather than cosmetic.

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Surviving Sepsis — Practice Guidelines, Marketing Campaigns, and Eli Lilly

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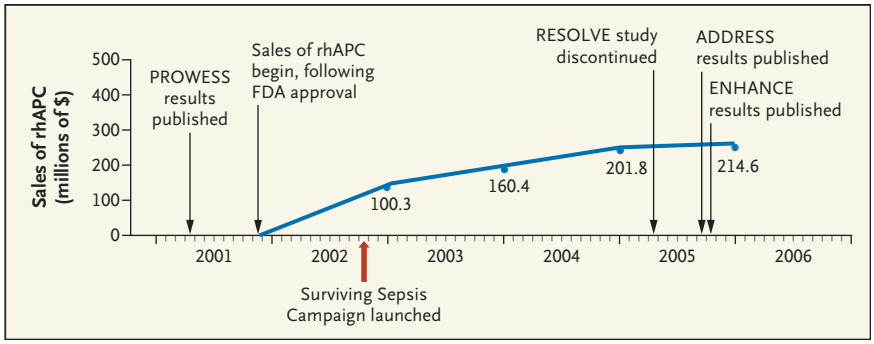
Practice guidelines approved by expert panels are intended to standardize care in such a way as to improve health outcomes. In recent years, the developers of such standards have started grouping evidence-based interventions into "bundles," on the theory that inducing physicians to follow multiple recommendations written into a single protocol has a measurable effect on patients' outcomes. As a side effect, bundled performance measures are ready-made for use in pay-for-performance initiatives, which can base reimbursement on compliance with all the components.

Unfortunately, the development of such clusters is vulnerable to manipulation for inappropriate — and possibly harmful — ends. Seeing in these bundles a potentially powerful vehicle for promoting their products, pharmaceutical and medical-device companies have begun to invest in influencing the adoption of guidelines that serve their own financial goals. A case in point is the development of guidelines for the treatment of sepsis, which was orchestrated as an extension of a pharmaceutical marketing campaign.[1,2] Although its advocates viewed this effort as an important approach to reducing sepsis-related mortality, the campaign appears to have usurped guideline development for commercial purposes, possibly compromising highly regarded, third-party arbiters of medical quality in the process. Such intrusion into an initiative to benefit public health is of particular concern in this instance, since the drug incorporated into the performance measures was endorsed on the basis of a single controversial phase 3 trial that was still being called into question by additional studies even as the committee did its work.

In 2001, the Food and Drug Administration (FDA) approved Eli Lilly's Xigris (recombinant human activated protein C, or rhAPC, also known as drotrecogin alfa [activated]) for the treatment of sepsis. This approval was based primarily on a single phase 3 randomized, controlled trial — the Recombinant Activated Human Protein C Worldwide Evaluation in Severe Sepsis (PROWESS) study, published the same year — which showed a significant overall survival benefit at 28 days. The FDA acknowledged that there was controversy surrounding this decision, and half the members of the agency's advisory panel, pointing to methodologic and other important problems with the PROWESS study, voted to require that a confirmatory trial be performed before approval was granted. In its approval statement, the FDA recommended using rhAPC in patients deemed, on the basis of an Acute Physiology and Chronic Health Evaluation II score of 25 or more, to have a particularly high risk of death; since this criterion had not been prospectively validated, the agency asked Lilly to perform additional testing in selected subgroups. In the face of such uncertainty, initial sales of rhAPC fell short of market expectations (see timeline).[3]

To improve sales of rhAPC, in 2002, Lilly hired Belsito and Company, a public relations firm, to develop and help implement a three-pronged marketing strate-

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Timeline of Controlled Trials of rhAPC, Regulatory Actions, Yearly Sales, and the Marketing Initiative by Eli Lilly.

The Surviving Sepsis Campaign consisted of three phases — an initial one defining the need to treat sepsis, a second one developing treatment guidelines, and a third one developing and implementing performance bundles based on the guidelines. The four data points show end-of-year sales. The company had predicted annual sales of $300 million to $500 million.

gy.[1] First, the product's sales were to be supported by marketing initiatives targeted to physicians and the medical trade media.[1] Second, because rhAPC was relatively expensive, word would be spread that the drug was being rationed and physicians were being "systematically forced" to decide who would live and who would die.[1,3] As part of this effort, Lilly provided a group of physicians and bioethicists with a $1.8 million grant to form the Values, Ethics, and Rationing in Critical Care (VERICC) Task Force, purportedly to address ethical issues raised by rationing in the intensive care unit.[3] Finally, the Surviving Sepsis Campaign was established, in theory to raise awareness of severe sepsis and generate momentum toward the development of treatment guidelines.

The first phase of the Surviving Sepsis Campaign was introduced at an October 2002 meeting of the European Society of Intensive Care Medicine (ESICM). In the second phase, launched in June 2003, international experts in critical care and infectious diseases were convened to create guidelines for sepsis management, which were published in *Critical Care Medicine* in March 2004.[4] Lilly provided more than 90% of the funding for these two phases, and many participants had financial or other relationships with the company.[1,4] According to the Council of Public Relations Firms, Belsito helped to assemble the VERICC Task Force and launch the campaign, and initiated a media-outreach program to "raise awareness" of alleged rationing in severe sepsis with the intent of generating demand for rhAPC.[1]

Campaign participants might argue that, regardless of Lilly's concerted efforts, the guidelines were not influenced by the company and represent best practice based on the evidence that was available — largely from randomized, controlled trials.[4] Although such trials represent the gold standard of medical evidence, overreliance on them in the construction of guidelines has a tendency to favor new drugs and devices, which typically undergo at least one such trial in order to obtain government approval. In this instance, that reliance meant that rhAPC was given a highly favorable rating (grade B), whereas established therapies for sepsis (such as antibiotics, fluids, and vasopressors), though included in the recommendations, received lower ratings (grade D or E), because most had not undergone randomized, controlled trials owing to a lack of equipoise.

This imbalance is made more troubling by the campaign's failure to discuss persisting concern about rhAPC, which has been reinforced by recent trials. After the PROWESS study, which had demonstrated an increased risk of serious bleeding, two other controlled trials — the Administration of Drotrecogin Alfa (Activated) in Early Stage Severe Sepsis (ADDRESS) study and the Resolution of Organ Failure in Pediatric Patients with Severe Sepsis (RESOLVE) study — both of which were terminated early because they were deemed unlikely to show a significant difference in their primary end points, confirmed that increase in risk and resulted in warnings submitted by Lilly to the FDA regarding the use of rhAPC. Although the results of the ADDRESS study were reported at the October 2004 ESICM meeting, no mention of the study was included in a supplement to the Surviving Sepsis Campaign Guidelines published the following month in *Critical Care Medicine*. Results from one open-label trial, the Extended Evaluation of Recombinant Human Activated Protein C (ENHANCE) study, published in October 2005, indicated that the risk of bleeding associated with rhAPC might actually be greater than originally esti-

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mated. Although data from the ENHANCE trial were available and are included in the guideline supplement, the possible magnitude of this increased risk (a 28-day incidence of serious bleeding of 6.5%, as compared with 3.5% in the PROWESS study) is not noted. Moreover, the efficacy of rhAPC has not been prospectively demonstrated in the patient population for which the drug is currently recommended.

Eleven professional societies are cited as sponsors of the Surviving Sepsis Campaign Guidelines. The Infectious Diseases Society of America (IDSA), however, declined to endorse them. According to Naomi O'Grady, the physician who chaired the IDSA's Standards and Practice Guidelines Committee from 2002 to 2005, the organization found fault with the manner in which the guidelines were developed, the use of a suboptimal rating system, and their sponsorship by a drug company. The peer-review process conducted by the IDSA might provide a model for an objective system of rating proposed guidelines in the future. But in this case, even the fact that the society decided not to endorse the recommendations is not widely known. According to Dante L. Landucci, an intensivist at East Carolina University, *Critical Care Medicine*, which published the guidelines, removed mention of the IDSA's rejection from his invited editorial on the subject that appeared in print 3 months after the guidelines did.

As part of the third phase of the campaign, Lilly awarded unrestricted grants for an "Implementing the Surviving Sepsis Campaign" program.[5] The main goal of this phase, launched in mid-2004, is the creation of performance bundles based on selected recommendations from the campaign guidelines. Again, many participants have self-reported financial or other relationships with Lilly.[4,5] Despite the persisting scientific controversy surrounding its safety and efficacy, rhAPC is included in one of these performance bundles. Neither the campaign's manual on bundle implementation nor a cover letter from the president of the Society of Critical Care Medicine mentions the ADDRESS and RESOLVE trials or the warnings they precipitated.[5] In formulating and promoting the bundles, the campaign sought to collaborate with public, not-for-profit arbiters of the quality of health care, including the Voluntary Hospital Association, the Institute for Healthcare Improvement, and the Joint Commission on Accreditation of Healthcare Organizations.[2,5]

Implementation of the bundles is being advocated nationally in workshops organized under the auspices of the Society of Critical Care Medicine and funded by Lilly. Furthermore, the campaign has lobbied state governments to adopt the bundles. Efforts to institute these measures internationally are being promoted in a program called the "Surviving Sepsis Campaign Roadshow," also subsidized by Lilly. In addition, the company funds *Advances in Sepsis*, a widely distributed periodical that publicizes the campaign. These activities continue unabated amid increasing calls for a new, prospective study of rhAPC.

When properly formulated and applied, practice guidelines and performance standards hold the promise of improving patients' outcomes. Professional societies and other stakeholders must work together to promote a consistent guideline-development process, a robust rating system for guidelines that is applicable to all subspecialties, and a policy that prohibits the pharmaceutical and medical-device industries from directly or indirectly funding or influencing practice standards. The challenges involved in producing first-rate guidelines and performance standards are only exacerbated by the intrusion of marketing strategies masquerading as evidence-based medicine.

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